FINM3013 Introduction to Financial Derivatives

Chapter 04

Class exercises

Multiple Choice Questions

1. An interest rate is 5% per annum with continuous compounding. What is the equivalent rate with semiannual compounding?
   A. 5.06%
   B. 5.03%
   C. 4.97%
   D. 4.94%

2. An interest rate is 12% per annum with semiannual compounding. What is the equivalent rate with quarterly compounding?
   A. 11.83%
   B. 11.66%
   C. 11.77%
   D. 11.92%

3. The two-year zero rate is 6% and the three year zero rate is 6.5%. What is the forward rate for the third year? All rates are continuously compounded.
   A. 6.75%
   B. 7.0%
   C. 7.25%
   D. 7.5%

4. Which of following describes forward rates?
   A. Interest rates implied by current zero rates for future periods of time
   B. Interest rate earned on an investment that starts today and last for n-years in the future without coupons
   C. The coupon rate that causes a bond price to equal its par (or principal) value
   D. A single discount rate that gives the value of a bond equal to its market price when applied to all cash flows

5. The six month and one-year rates are 3% and 4% per annum with semiannual compounding. Which of the following is closest to the one-year par yield expressed with semiannual compounding?
   A. 3.99%
   B. 3.98%
   C. 3.97%
   D. 3.96%